Convex function

1) $f(x) = x^2,\ x \in R$

$\nabla f = 2x \Rightarrow \nabla'' f = 2 > 0 \Rightarrow f(x)$ is convex

2) $f(x) = e^{x^2},\ x \in R$

$u = x^2$

$\Rightarrow \nabla f = u' e^u = 2x e^{x^2}$

$u_1 = 2x \quad v = e^{x^2} \Rightarrow u' = 2 \quad v' = 2x e^{x^2}$

$\Rightarrow \nabla'' f = u_1' v + v' u_1 = 2e^{x^2} + (2x)^2 e^{x^2} > 0$

$\forall x \in R \Rightarrow f(x)$ is convex

3) $f(x,y) = x^2 + 3xy + 2y^2 \quad x \in R, y \in R$

$\Rightarrow \nabla f(x,y) = \begin{bmatrix} 2x + 3y \\ 3x + 4y \end{bmatrix}$

$\Rightarrow \nabla'' f(x,y) = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$

$\det(\nabla'' f) = 2.4 - 3.3 = -1 < 0$

$\Rightarrow$ not convex

Log sum exp

$f(x) = \log(e^{x_1} + \dots + e^{x_n}) = \log(\sum^n e^{x_n})$

$\log(u) = \frac{u'}{u} \Rightarrow \nabla f = \begin{bmatrix} \frac{e^{x_1}}{\sum e^{x_n}} \\ \vdots \\ \frac{e^{x_n}}{\sum e^{x_n}} \end{bmatrix}$

$\Rightarrow \nabla'' f$ has 2 cases $H_{ii}\ (i=j)$

$H_{ij}\ (i \neq j)$

We have $u = e^{x_i} \quad v = \sum^n e^{x_n}$

$H_{ii} \Rightarrow u' = e^{x_i} \quad v' = e^{x_i}$

$\Rightarrow H_{ii} = \frac{u'v - v'u}{v^2}$

$= \frac{e^{x_i}}{\sum^n e^{x_n}} - \left(\frac{e^{x_i}}{\sum^n e^{x_n}}\right)^2$

$H_{ij} \Rightarrow u' = 0 \quad v' = e^{x_j}$

$\Rightarrow H_{ij} = -\frac{e^{x_i}}{\sum^n e^{x_n}} \cdot \frac{e^{x_j}}{\sum^n e^{x_n}}$

Define: $z_j = \frac{e^{x_j}}{\sum^n e^{x_n}} \quad z_i = \frac{e^{x_i}}{\sum e^{x_n}}$

$\Rightarrow H_{ii} = z_i - z_i^2 \qquad H_{ij} = -z_i . z_j$

$\Rightarrow H = \nabla^2 f(x) = diag(z) - z^T z \quad [z = (z_1, z_2 \dots z_n)]$

$(H_{ij} = diag(z)^{ij} - (z^T z)^{ij} = 0 - z_i z_j$

$H_{ii} = diag(z)^{ii} - (z^T z)^{ii} = z_i - z_i^2)$

$\Rightarrow v^T H v = v^T (diag(z) - z^T z) v$

$= (v^T diag(z) v) - v^T z^T z v$

$= \underset{>0}{v^2 \sum^n z_i} - \underset{>0}{(vz)^2} \Rightarrow f(x)$ convex

Geometric mean

$f(x) = (\prod_{i=1}^n x_i)^{\frac{1}{n}}$

We have $u = e^{\log(u)}$

$\Rightarrow f(x) = e^{\log[(\prod^n x_i)^{\frac{1}{n}}]}$

$\Rightarrow f(x) = g(h(x))$

$\Rightarrow g(u) = e^u$ is convex $\underline{\underline{(1)}}$ with $\nabla'' g = e^u$

$h(x) = \log[(\prod_1^n x_i)^{\frac{1}{n}}] = \frac{1}{n} \sum_1^n \log x_i$

$\nabla h(x) = \frac{1}{n} \begin{bmatrix} \frac{1}{x_1} \\ \vdots \\ \frac{1}{x_n} \end{bmatrix} \Rightarrow \nabla'' h(x) = -\frac{1}{n} \begin{bmatrix} \frac{1}{x_1^2} \\ \vdots \\ \frac{1}{x_n^2} \end{bmatrix}$

$f'(x) = g'(h(x)) . h'(x)$

$u = g'(h(x)) \quad v = h'(x)$

$\Rightarrow u' = g''(h(x)) . h'(x) \quad v' = h''(x)$

$f''(x) = u'v + v'u = g''(h(x))(h'(x))^2 + h''(x) g'(h(x))$

$= e^u \left(\frac{1}{n} \begin{bmatrix} \frac{1}{x_1} \\ \vdots \\ \frac{1}{x_n} \end{bmatrix}\right)^2 - \frac{1}{n} \begin{bmatrix} \frac{1}{x_1^2} \\ \vdots \\ \frac{1}{x_n^2} \end{bmatrix} . e^u$

$= \underset{>0}{e^u} \left(\underset{<0}{\frac{1}{n^2} - \frac{1}{n}}\right) . \underset{>0}{\begin{bmatrix} \frac{1}{x_1^2} \\ \vdots \\ \frac{1}{x_n^2} \end{bmatrix}} \leq 0$

$\Rightarrow f(x)$ concave